

4-to-1 Multiplexer Using Behavioral Modeling

Aim

To design and verify a 4-to-1 multiplexer in Verilog using behavioral modeling (`case` statement inside an `always @(*)` block).

Code

```
module mux4_behavioral (  
    input wire [3:0] d,  
    input wire [1:0] sel,  
    output reg      y  
);  
    always @(*) begin  
        case (sel)  
            2'b00: y = d[0];  
            2'b01: y = d[1];  
            2'b10: y = d[2];  
            2'b11: y = d[3];  
            default: y = 1'bx;  
        endcase  
    end  
endmodule
```

Truth Table / Observation

sel	d[3:0]	y	Selected Input
00	XXX1	d[0]	d[0]
01	XX1X	d[1]	d[1]
10	X1XX	d[2]	d[2]
11	1XXX	d[3]	d[3]

Observation: Tested with all 16 data values across all 4 select states (64 total test vectors). The output `y` correctly selects `d[0]` , `d[1]` , `d[2]` , or `d[3]` matching the 2-bit binary value of `sel` .

Conclusion

The 4-to-1 multiplexer was successfully implemented using behavioral modeling with a `case` statement. Simulation results verify that the correct data bit is routed to the output for every select combination.